

Publishable Research Package: Regional Electricity Demand and Scope 2 Carbon Scenarios for Indian Data Centres

Executive summary

I have rebuilt the original seven-page draft into a substantially different, reproducible journal manuscript rather than simply polishing its language. The supplied paper contained a strong research idea, but its original regional emissions table, full-load assumption, renewable-energy treatment, city-specific grid factors and several source claims would have created serious peer-review problems. The original file also carries “© 2025 IEEE | Proceedings of IEEE ICECET 2025” branding, which creates an important prior-publication issue discussed below. [filecite turn0file0](#)

The revised paper is titled:

**Regional electricity demand and Scope 2 carbon scenarios for Indian data centres:
an uncertainty-aware assessment of seven major hubs**

The reconstructed manuscript contains approximately **6,430 words from the Abstract through the References**, keeping it below IJSEPM's current **7,000-word maximum including everything**. It uses British English, a single-column layout, page and line numbering, numbered references, embedded figures and the article structure expected by the journal. IJSEPM allows flexible formatting at initial submission but specifically requires Word format, no columns, page and line numbering and no more than 7,000 words. ¹

The central analytical result is very different from the original draft. For the seven reported H1 2026 hubs, the model covers **1,755 MW of operational IT capacity**. With a transparent central assumption of **65% IT-load factor**, **PUE 1.45** and CEA's latest **0.675 tCO₂/MWh FY2025–26 weighted-average Indian-grid factor**, estimated annual facility electricity consumption is **14.49 TWh** and location-based Scope 2 emissions are **9.78 MtCO₂/year**. CEA explicitly reports India as one unified grid and gives 0.675 tCO₂/MWh as its FY2025–26 weighted-average factor. ²

The full package is available below.

Deliverable	Download
Complete package containing everything	Download complete ZIP
Journal manuscript (.docx)	Download IJSEPM-ready Word manuscript
City-level dataset (.csv)	Download city dataset
Reproducible Python notebook (.ipynb)	Download executed notebook
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Deliverable	Download
All scenario outputs (.csv)	Download scenario results
Monte Carlo output (.csv)	Download Monte Carlo summary
Location-based sensitivity (.csv)	Download sensitivity results
Market-based sensitivity (.csv)	Download sensitivity results
Methodology Mermaid source	Download methodology diagram
Timeline Mermaid source	Download evidence timeline
Package instructions	Download README

Research reconstruction and current evidence base

The paper now uses **H1 2026** rather than mixing 2024, 2025 and projected capacity values. Cushman & Wakefield's official September 2026 update reports that Indian operational data-centre capacity had reached **nearly 1.8 GW**, with a development pipeline of nearly **3.9 GW**, and identifies Mumbai as the country's largest market. ³

Detailed figures reported from the same H1 2026 update are:

Hub	Operational IT capacity used in model
Mumbai	890 MW
Chennai	214 MW
Delhi-NCR	183 MW
Bengaluru	162 MW
Hyderabad	151 MW
Pune	133 MW
Kolkata	22 MW
Modelled total	1,755 MW

Those detailed city figures are reproduced in reporting on the Cushman & Wakefield study, which gives a headline top-seven operational total of **1,789 MW**. ⁴ The city values therefore fall **34 MW, or about 1.9%, below the headline total**. I deliberately did **not** rescale or manipulate the city numbers to make them add up. Instead, the discrepancy is disclosed in the manuscript and capacity uncertainty is included in the Monte Carlo analysis.

This is much stronger than the original approach, where market shares assigned to cities added to more than 100%.                                     

The wider electricity-demand context is also corrected. TERI stated in September 2025 that Indian data centres represented **less than 1% of national electricity demand at that time**, while potentially reaching as much as **6% by 2030** under strong growth. That replaces the unsupported blanket

statement in the draft that the sector already represented about 2% of Indian electricity consumption. ⁵

Internationally, the IEA estimates that data centres consumed around **415 TWh, or 1.5% of global electricity, in 2024**. Its 2026 update projects approximately **485 TWh in 2025 and 950 TWh in 2030**, while explicitly emphasising uncertainty in AI deployment, infrastructure and power-system constraints.

⁶ That is why the manuscript treats future development as uncertain rather than presenting a single 2030 capacity number as fact.

The literature review has also been strengthened around peer-reviewed and institutional sources. Masanet et al.'s *Science* study demonstrated why workload growth and electricity demand cannot simply be extrapolated proportionally when server and facility efficiencies are changing, while the latest LBNL work uses bottom-up equipment, facility and cooling assumptions and explicitly reports scenario ranges. ⁷

Carbon-accounting model and research results

The largest methodological change is that the manuscript no longer treats installed megawatts as annual electricity.

For each hub, the model calculates:

$$E_{IT} = C_{IT} \times LF \times 8760$$

and

$$E_{facility} = E_{IT} \times PUE$$

where C_{IT} is operational IT capacity and LF is the average IT-load factor.

The central scenario uses:

Parameter	Central value	Treatment
IT-load factor	0.65	Analytical assumption
PUE	1.45	Scenario assumption informed by contemporary PUE evidence
Grid emission factor	0.675 tCO ₂ /MWh	CEA FY2025–26
Qualifying renewable procurement	40%	Market-based scenario variable only

The PUE assumptions are deliberately more defensible than the original city-specific ranges. Uptime Institute's 2026 global survey reports an ordinary annual-average PUE of **1.52**, a capacity-weighted PUE of **1.36**, and notes that newer large facilities routinely report **1.3 or below**. ⁸ There is not enough independently verified public evidence to claim, for example, that all Delhi-NCR data centres have one average PUE while Bengaluru has another. The revised paper therefore avoids false regional precision.

Central city results

Under the central scenario, the model produces:

Hub	Facility electricity	Location-based Scope 2
Mumbai	7.348 TWh/yr	4.960 MtCO ₂ /yr
Chennai	1.767 TWh/yr	1.193 MtCO ₂ /yr
Delhi-NCR	1.511 TWh/yr	1.020 MtCO ₂ /yr
Bengaluru	1.338 TWh/yr	0.903 MtCO ₂ /yr
Hyderabad	1.247 TWh/yr	0.842 MtCO ₂ /yr
Pune	1.098 TWh/yr	0.741 MtCO ₂ /yr
Kolkata	0.182 TWh/yr	0.123 MtCO ₂ /yr
Total	14.490 TWh/yr	9.781 MtCO₂/yr

These should be understood as **scenario estimates**, not operator-reported emissions.

The correction is substantial. CEA states that the Indian grid has been synchronised as a single grid since December 2013 and reports a FY2025–26 weighted-average emission factor of **0.675 tCO₂/MWh**, with a simple operating margin of **0.963**, build margin of **0.446** and combined margin of **0.705 tCO₂/MWh**.² Consequently, the manuscript no longer claims that CEA provides different city factors such as 0.82 for Delhi and 0.58 for Bengaluru.

This also provides a powerful plausibility check against the original draft. Even **890 MW** of Mumbai IT capacity running continuously at **100% utilisation** with **PUE 1.60** would consume about 12.47 TWh/year; using the current CEA average factor gives only about **8.42 MtCO₂/year**, rather than the draft's approximately **75 MtCO₂/year**. The previous regional table was therefore not salvageable by wording edits alone. [filecite turn0file0](#)

Location- versus market-based Scope 2

Renewable procurement is now treated according to Scope 2 accounting principles instead of multiplying location-based emissions by $(1 - \text{renewable share})$.

GHG Protocol distinguishes a **location-based method**, reflecting average emissions from the grid serving consumption, from a **market-based method**, which represents qualifying electricity products and contractual instruments. Where relevant, companies report both.⁹

Therefore:

$$Scope2_{LB} = E_{facility} \times EF_{grid}$$

and renewable PPAs or certificates **do not reduce this figure**.

For market-based scenarios:

$$Scope2_{MB} = E_{unmatched} \times EF_{unmatched}$$

with matched qualifying renewable electricity treated according to the contractual instrument.

This matters because the current CEA weighted-average factor already includes renewable generation. CEA explicitly states that the weighted average incorporates renewable generation into the Indian-grid average. ² Reducing it again by a general renewable percentage would effectively count part of the renewable benefit twice.

For the **40% renewable central market-based scenario**, the paper therefore reports two transparent bounds:

Grid-average fallback: 5.87 MtCO₂/year

CEA operating-margin proxy sensitivity: 8.37 MtCO₂/year

The second number uses CEA's 0.963 tCO₂/MWh simple operating margin only as a deliberately conservative sensitivity. It is **not described as India's residual mix**.

This distinction becomes increasingly important because the GHG Protocol's proposed Scope 2 revisions would strengthen geographic deliverability, temporal matching and residual-mix treatment and propose fossil-only fallback where a suitable residual mix does not exist. Those changes remain proposals rather than the current final standard. ¹⁰

Indian operators are already demonstrating that high contractual renewable shares are possible. Nxtra stated in 2024 that additional wind and solar procurement would raise renewable electricity to approximately **70% of its core data-centre energy mix**; in 2025 it announced a further **205,167 MWh** of solar-wind hybrid supply. ¹¹ The revised paper uses this evidence to justify the *range* of the renewable scenario, not to assume every Indian facility has achieved 70%.

Uncertainty, figures and reproducibility

The deterministic paper has been upgraded to a **50,000-draw Monte Carlo model**. The notebook varies capacity, IT-load factor, PUE, grid intensity and qualifying renewable share independently.

The ranges used are deliberately transparent:

Variable	Minimum	Central/mode	Maximum
Capacity multiplier	0.95	1.00	1.05
IT-load factor	0.50	0.65	0.80
PUE	1.35	1.45	1.60
Grid factor	0.64125	0.67500	0.70875 tCO ₂ /MWh
Qualifying renewable share	0%	40%	70%

These are explicitly identified as modelling distributions, **not measured statistical probability distributions**.

The resulting uncertainty estimates are:

Output	Mean	5th percentile	95th percentile
Facility electricity	14.66 TWh/yr	12.20	17.19
Location-based Scope 2	9.89 MtCO ₂ /yr	8.21	11.64
Market-based, grid fallback	6.26 MtCO ₂ /yr	3.84	9.02
Market-based, OM proxy	8.93 MtCO ₂ /yr	5.49	12.85

The rank-correlation sensitivity analysis shows that **IT-load factor is by far the largest uncertainty in location-based emissions** ($\rho \approx 0.90$). PUE follows at about 0.32. For the market-based calculation, **renewable procurement share dominates** ($\rho \approx -0.91$). This is analytically important: obtaining real IT utilisation data is likely to improve current electricity estimates more than attempting to fine-tune an assumed city-specific PUE.

The notebook is already executed with a fixed random seed of **42**, so another researcher can rerun the study and obtain the same Monte Carlo output.

The package also contains eight figures in both **300-dpi PNG and SVG vector format**, including the India hub map, capacity comparison, city emissions chart, scenario comparison, Monte Carlo distribution, sensitivity chart, methodology flowchart and evidence timeline. IJSEPM specifically prefers vector graphics and requires descriptive axes and embedded manuscript figures, so SVG copies are included alongside the Word-embedded PNG versions. ¹

Key individual files include:

[India hub map – SVG](#)

[Scenario emissions – SVG](#)

[Monte Carlo distribution – SVG](#)

[Sensitivity analysis – SVG](#)

[Methodology flowchart – SVG](#)

[Evidence timeline – SVG](#)

Journal fit and submission status

I retained **International Journal of Sustainable Energy Planning and Management (IJSEPM)** as the primary target rather than switching journals.

That decision is based on its current scope: IJSEPM explicitly covers **energy-system analysis, feasibility studies and public regulation**, including regional scenario/model analyses of sustainable-energy transitions. It is published by Aalborg University Open Publishing and states that it provides **free access for readers and authors**. ¹²

That makes it unusually suitable for your goal: a legitimate journal with **no author publication charge**, while the reconstructed paper now has a much stronger energy-planning and policy-analysis orientation.

The current manuscript also follows several specific IJSEPM requirements:

- it is in **Word** and **single column**;
- the principal journal text is approximately **6,430 words**, below the 7,000-word maximum;
- it uses **British English**;
- references are **numbered in square brackets**;
- reference entries contain URLs or DOI links;
- it has page and continuous line numbering;
- the Abstract is below the journal's **200-word maximum**;
- tables and graphics are embedded;
- a reproducible data-availability statement is included.

Those requirements come directly from the current journal author instructions. ¹

One practical consideration is that IJSEPM currently warns that it **intermittently closes to new submissions** because of high numbers of out-of-scope manuscripts. ¹² The journal should therefore be checked at the moment you are ready to upload, rather than assuming the submission portal will always be open.

Critical checks before actual submission

The package is **research- and formatting-ready**, but there is one issue that must be resolved before pressing “Submit”.

Your supplied document visibly states “© 2025 IEEE | Proceedings of IEEE ICECET 2025” and has an IEEE proceedings footer on multiple pages. ¹ IJSEPM's current submission checklist states that an ordinary submitted article must **not have been published before**, while noting special rules for conference special issues. ¹

Therefore, determine which of these situations applies:

If the draft was never actually published by IEEE or ICECET: remove all IEEE/proceedings branding and submit the reconstructed paper as the original journal manuscript.

If an IEEE conference paper was actually published: do not submit the journal version without disclosure. The editor needs to know the conference citation, and the journal version must satisfy any permitted extension/originality requirements. The present manuscript is methodologically much more substantial than the seven-page draft—new temporal data, corrected carbon accounting, load-factor modelling, dual Scope 2 accounting, Monte Carlo analysis, sensitivity analysis, new tables, new figures and reproducible data/code—but that methodological extension does not eliminate the requirement to disclose a published predecessor. ¹

The Word document contains an **author-facing pre-submission note and Executive summary** at the front so that this warning is not missed. Those two author-facing elements should be deleted before the actual journal upload. The manuscript proper begins with the title/Abstract and remains within the journal's word limit after that removal.

The other final metadata item is the affiliation. IJSEPM asks for a complete institutional affiliation including a **street address and postal code**; the available draft only gives “Computer Department, Zeal Institution, Pune, India”, so the exact institutional name and postal address should be verified before

upload. The journal also asks authors to propose **two to five independent reviewers** in the submission system. ¹

The core research package itself is complete:

[Download the complete publishable research package](#)

- ¹ <https://journals.aau.dk/index.php/sepm/about/submissions>
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<https://economictimes.indiatimes.com/industry/services/property/-cstruction/indias-data-centre-capacity-set-to-surge-by-3860-mw-by-2030-fueling-digital-growth-and-investment-opportunities/articleshow/134283064.cms?from=mdr>
- ⁵ <https://www.teriin.org/press-release/greening-indias-digital-future-teri-and-nsefi-sign-pact-sustainable-data-centers>
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- ⁷ <https://doi.org/10.1126/science.aba3758>
<https://doi.org/10.1126/science.aba3758>
- ⁸ <https://intelligence.uptimeinstitute.com/resource/growing-pue-advantage-larger-data-centers>
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- ⁹ <https://ghgprotocol.org/blog/interim-update-accounting-biomethane-certificates>
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- ¹⁰ <https://ghgprotocol.org/blog/upcoming-scope-2-public-consultation-overview-revisions>
<https://ghgprotocol.org/blog/upcoming-scope-2-public-consultation-overview-revisions>
- ¹¹ <https://www.airtel.in/press-release/02-2024/nxtra-to-procure-140208-mwh-renewable-energy-for-its-data-centres>
<https://www.airtel.in/press-release/02-2024/nxtra-to-procure-140208-mwh-renewable-energy-for-its-data-centres>
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